12) The oblique asymptote of $f(x) = \frac{2x^{2}+5}{x+2}$ is

A) $y = 2x+2$ B) $y = 2x+7$ C) $y = 2x-4$ D) $y = 2x-1$

13) The minimum value of $f(x) = x^{2} - 2x + 3$ on $[0,3]$ is equal to

A) 1 B) 3 C) 2 D) 6

14) $\int \frac{dx}{11x^{2}+7} =$

A) $\frac{1}{\sqrt{7}} \tan^{-1}\left(\sqrt{\frac{11}{7}}\,x\right) + c$

C) $\frac{1}{\sqrt{7}} \tan^{-1}\left(\sqrt{\frac{7}{11}}\,x\right) + c$

B) $\frac{1}{\sqrt{77}} \tan^{-1}\left(\sqrt{\frac{11}{7}}\,x\right) + c$

D) $\frac{1}{\sqrt{77}} \tan^{-1}\left(\sqrt{\frac{7}{11}}\,x\right) + c$

15) The slope of the tangent to the curve of $\sin^{-1} y = \ln(x^{2}+y)$ at the point $(3,0)$ is equal to

A) $\frac{3}{4}$ B) $\frac{4}{3}$ C) $\frac{2}{3}$ D) $\frac{3}{2}$

16) $\int_{\sqrt[3]{\ln 2}}^{\sqrt[3]{\ln 11}} e^{x^{3}+2\ln x} =$

A) 3 B) 1 C) 0 D) none

17) $\int_{0}^{\frac{\pi}{4}} \sec x \, dx =$

A) $\ln\left(\sqrt{2}+1\right)$ B) $\ln\left(\sqrt{3}+1\right)$ C) $\ln\left(\sqrt{3}+2\right)$ D) $\ln\left(\sqrt{2}+3\right)$

18) If f is continuous on $[1,10]$, $F(x) = \int_{c}^{x} f(t)\,dt$, $F(3) = 5$ and $F(6) = 11$ then $\int_{3}^{6} f(x)\,dx =$

A) 5 B) 9 C) 6 D) 4

19) $\int \frac{dx}{3e^{-x}+1} =$

A) $\ln(1+e^{x})+c$ B) $\ln(3+e^{x})+c$ C) $\ln(3+e^{-x})+c$ D) $\ln(1+e^{-x})+c$

20) The area of the region bounded by $y = \frac{1}{9+x^{2}}$, $y = -\sqrt{9-x^{2}}$ and $x = 0$, $x = 3$

(in square units) is equal to

A) $\frac{7}{3}\pi$ B) $\frac{3}{7}\pi$ C) $\frac{6}{7}\pi$ D) $\frac{7}{6}\pi$

Part One- Multiple Choice

40 Points

1) The domain of $f(x)=\frac{1}{\sqrt{x-1}-3}$

A) $[1,\infty)$ [✓ B) $[1,\infty)\setminus\{10\}$] C) $[1,\infty)\setminus\{3\}$ D) $[1,\infty)\setminus\{9\}$

2) The range of the function $f(x)=|\sin x|+6$ is

[✓ A) $[6,7]$] B) $[5,7]$ C) $[-1,1]$ D) none

3) $\lim\limits_{x\to-\infty}\frac{x^4+x+1}{-x+2}=$

A) 0 B) -1 C) $-\infty$ D) ∞

4) If $\lim\limits_{x\to2}\left(\frac{1}{x-3}-\frac{a}{x^2-9}\right)=b$ where $a,b\in\mathbb{R}$ then the order pair $(a,b)=$

A) $(6,\frac{1}{3})$ B) $(\frac{1}{3},6)$ C) $(6,\frac{1}{6})$ D) $(\frac{1}{6},6)$

5) $\lim\limits_{x\to\infty}\frac{\sqrt{4x^2+x+1}+5x}{-x+1}=$

A) 7 B) -7 C) ∞ [✓ D) $-\infty$]

6) Given that $f(x)=\frac{x^2}{e^{2x}-5}$ then f has discontinuity at $x=$

A) $\ln 5$ B) $\ln\sqrt{5}$ C) $2\ln 5$ [✓ D) $\ln\left(\frac{5}{2}\right)$]

7) If $f(x)=\begin{cases} x\sin\left(\frac{1}{x}\right) & , x\neq 0 \\ 0 & , x=0 \end{cases}$ then $f'(0)=$

A) 1 B) -1 C) 0 D) does not exist

8) Given that $f'(x)=6xf(x)$, $f\left(\frac{\pi}{6}\right)=\sqrt{27}$, $g(x)=f(\sin^{-1}x)$ then $g'\left(\frac{1}{2}\right)=$

A) 6π B) 6 C) 4π D) 4

9) At $x=5$ the function $f(x)=\sqrt[3]{(x-5)^2}$ has

A) a vertical tangent
B) a cusp
C) a relative maximum
D) an inflection point

10) The function $f(x)=x(x-4)^3$ is decreasing on

A) $(-\infty,\infty)$ B) $(0,\infty)$ [✓ C) $[1,\infty)$] D) $(-\infty,1]$

11) The function $f(x)=x|x-2|$ is concave down on

A) $(-\infty,\infty)$ B) $(0,\infty)$ [✓ C) $(2,\infty)$] D) $(-\infty,2)$

لجنة السنافر

1) $f(x) = \frac{1}{\sqrt{x-1}-3}$ $\qquad \sqrt{x-1}-3=0$

$x-1=9$ (B)

$D_f=[1,\infty)\setminus\{10\}$ $\qquad x=10$

2) $f(x)=|\sin x|+6$ $\qquad \underset{+6}{0}\le \underset{+6}{|\sin x|} \le \underset{+6}{1}$ (A)

$R(f)=[6,7]$

3) $\lim_{x\to-\infty}\left(\frac{x^4+x+1}{-x+2}\right)=\lim_{x\to-\infty}(-x^3)=\infty$ (D)

4) $\lim_{x\to3}\left(\frac{1}{x-3}-\frac{a}{x^2-9}\right)=\lim_{x\to3}\left(\frac{x^2-9-a(x-3)}{x-3\ (x^2-9)}\right)$

$=\lim_{x\to3}\left(\frac{\cancel{x-3}\,(x+3-a)}{\cancel{x-3}\,(x^2-9)}\right)\neq b$

$6-a=0$

$\boxed{a=6}$

$\lim_{x\to3}(x+3-\underset{\text{تعويض } a}{\underline{a}}) = \cancel{b\lim_{x\to3}\overset{0}{(x^2-9)}}$

$\lim_{x\to3}\left(\frac{x-3}{x^2-9}\right)=b=\lim_{x\to3}\left(\frac{1}{x+3}\right)=\frac{1}{6}$ (C)

5) $\lim_{x\to\infty}\left(\frac{\sqrt{4x^2+x+1}+5x}{-x+1}\right)=\lim_{x\to-\infty}\left(\frac{|2x|+5x}{-x}\right)=\frac{7x}{-x}=-7$ (B)

6) $f(x)=\frac{x^2}{e^{2x}-5}$, $\quad e^{2x}-5=0$, $\quad x=\ln\sqrt{5}$ (B)

7) $f(x)=\begin{cases} x\sin\left(\frac{1}{x}\right), & x\neq0 \\ 0, & x=0\end{cases}$

$\frac{\sin(1/x)}{1/x}$, $t=\frac{1}{x}$ $\quad x\to 0$, $t\to\infty$, $\lim_{x\to\infty}\left(\frac{\sin t}{t}\right)$

$-\frac{1}{t}\le\frac{\sin(t)}{t}\le\frac{1}{t}$ ناخذ limit لكل الأطراف عندما $t\longrightarrow\infty$

$0\le\lim_{t\to\infty}\left(\frac{\sin t}{t}\right)\le 0$ $\quad\therefore f'(0)=0$ Ⓒ

8) $f(\pi/6)=\sqrt{27}$, $g(x)=f(\sin^{-1}x)$

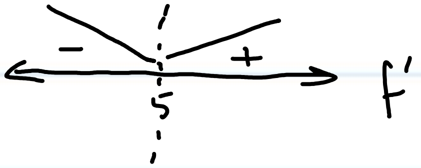
$f'(x)=6x\,f(x)$, $g'(1/2)=??$

$g'(x)=f'(\sin^{-1}x)\cdot\frac{1}{\sqrt{1-x^2}}$ | $f'(\pi/6)=\pi f(\pi/6)$

$=\sqrt{27}\,\pi$

$g'(x)=\frac{2}{\sqrt3}f'(\pi/6)=\frac{\cancel{\sqrt{27}}^{\,3}}{\cancel{\sqrt3}}\,2\pi=6\pi$ Ⓐ

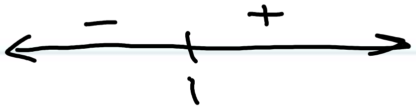
9) $f(x)=(x-5)^{2/3}$, $f'(x)=\frac{2}{3}(x-5)^{-1/3}$ Ⓑ



بما ان $f'(5)$ undifined اذن cusp رأس حاد

10) $f(x)=x(x-4)^3$, $f'(x)=3x(x-4)^2+(x-4)^3$

$f'(x)=\underbrace{(x-4)^2}_{+}(4x-4)$



$[-\infty,1]$ Ⓓ

11) $F(x)=\begin{cases}x^2-2x, & x\ge 2\\ 2x-x^2, & x<2\end{cases}$, $f'(x)=\begin{cases}2x-2\\ 2-2x\end{cases}$, $f''(x)=\begin{cases}2, & x\ge 0\\ \underline{-2,\ x<2}\end{cases}$

Ⓓ

12) $f(x)=\frac{2x^2+3}{x+2}$

$$\begin{array}{r} 2x-4 \\ x+2\,\overline{)\,2x^2+3} \\ -(2x^2+4x) \\ \hline -4x+3 \\ -(-4x-8) \\ \hline 11 \end{array}$$

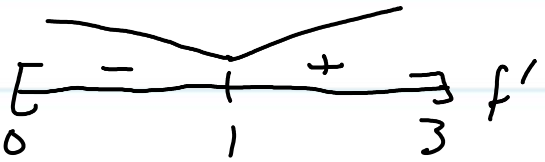
Ⓒ

Oblinqe

asymptote : $y = 2x-4$

13) $f(x) = x^2-2x+3$, $f'(x) = 2x-2 = 2(x-1)$

$f(1) = 2$ Ⓓ



14) $\int \frac{dx}{11x^2+7} = \frac{1}{\sqrt{77}}\tan^{-1}\left(\sqrt{\frac{11}{7}}\,x\right) + c$ Ⓑ

15) $\sin^{-1}y = \ln(x^2+y) \rightarrow \frac{y'}{\sqrt{1-y^2}} = \frac{2x+y'}{x^2+y}$ $(3,0)$

Ⓐ

$y' = \frac{6+y'}{9} \rightarrow 9y' = 6+y'$

$8y'=6 \rightarrow y' = \frac{3}{4}$

16) $\int_{\sqrt[3]{\ln 2}}^{\sqrt[3]{\ln 11}} e^{x^3+2\ln x}\,dx = \frac{1}{3}e^{x^3}\Big]_{\sqrt[3]{\ln 2}}^{\sqrt[3]{\ln 11}} = \frac{1}{3}(11-2) = 9/3 = 3$

Ⓐ

$* \; e^{x^3}\cdot e^{\ln x^2} = x^2e^{x^3}$

17) $\int_0^{\pi/4}\sec x\,dx = \ln|\sec x+\tan x|\Big]_0^{\pi/4} = \ln(\sqrt{2}+1) - \overset{0}{\cancel{\ln 1}}$

$\sec x = \frac{1}{\cos x}$ $\sec(\pi/4) = \sqrt{2}$ $\tan(0)=0$

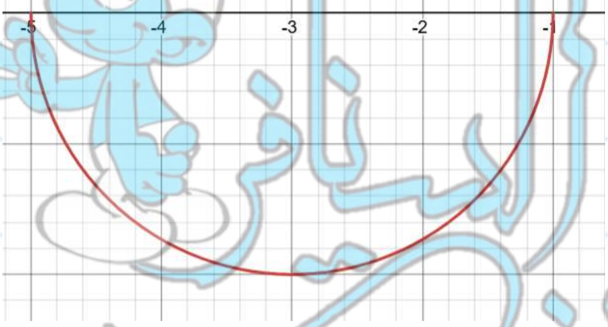
$\sec(0)=1$

18) $\int_3^6 f(x)\,dx = F(6) - F(3) = 11 - 5 = 6$ $\textcircled{C}$

19) $\int \frac{dx}{3e^{-x}+1} = \int \frac{dx}{e^{-x}(3+e^{x})} = \int \frac{e^{x}dx}{3+e^{x}} = \ln(3+e^{x}) + c$ $\textcircled{B}$

20) $\textcircled{A}$

• السؤال المقالي :-



المعطى : $f(x)$

$\downarrow$

المطلوب : $f'(x)$